

SYDNEY MOUNT ANNAN, Australia

WMO: 947570

Lat: 34.0614S

Lon: 150.7736E

Elev: 112

StdP: 99.99

Time Zone: 10.00 (AUS)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.5	2.7	-5.1	2.5	18.0	-3.0	3.0	16.2	8.1	16.4	6.8	16.0	0.6	190	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	35.4	20.3	33.0	20.2	30.8	19.7	23.0	29.8	22.2	28.2	21.5	27.1	3.9	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.1	24.7	20.5	15.4	24.1	19.9	14.8	23.7	68.6	30.0	65.5	28.1	63.1	27.0	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.6	5.0	-1.2	40.8	0.9	2.7	-1.8	42.8	-2.4	44.3	-2.9	45.9	-3.5	47.8		
(5)			-1.5	24.8	1.0	1.0	-2.2	25.5	-2.7	26.1	-3.3	26.7	-4.0	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	23.4	22.4	20.8	17.6	14.0	11.7	10.7	11.7	14.5	17.2	19.8	21.5		
	DBStd	5.04	3.13	2.77	2.48	2.54	2.22	1.89	1.95	2.04	2.52	3.07	3.14	3.13		
	HDD10.0	28	0	0	0	0	1	6	14	6	1	0	0	0		
	HDD18.3	1033	1	1	7	42	136	198	238	206	118	60	20	7		
	CDD10.0	2609	414	348	335	229	124	58	35	59	136	224	293	355		
	CDD18.3	572	157	115	83	21	1	0	0	0	4	26	63	103		
	CDH23.3	4851	1308	849	518	177	17	0	2	7	109	369	610	884		
	CDH26.7	1904	601	336	147	44	1	0	0	0	29	132	247	367		
(14) Wind	WSAvg	2.1	2.2	2.1	2.0	1.8	1.8	1.8	1.9	2.2	2.3	2.2	2.3	2.3		
(15) Precipitation	PrecAvg	803	85	79	79	78	56	73	34	58	41	74	81	64		
	PrecMax	1320	234	358	286	351	159	361	124	346	167	330	305	195		
	PrecMin	364	11	4	1	4	0	2	0	1	0	0	6	0		
	PrecStd	245	65	74	63	80	46	85	29	74	33	70	70	48		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	38.5	33.0	31.2	25.3	20.0	22.4	24.3	30.1	33.4	36.8	37.4		
		MCWB	20.7	20.4	20.1	17.4	14.9	13.0	12.9	13.2	15.7	17.0	18.4	19.7		
	2%	DB	35.4	33.5	30.1	27.2	23.1	18.7	19.3	21.6	26.3	30.3	32.3	33.6		
		MCWB	21.6	21.5	20.2	17.3	14.5	12.8	11.2	12.4	14.3	16.6	18.8	19.7		
	5%	DB	32.5	30.3	28.0	25.1	21.3	17.6	17.8	19.6	23.6	27.6	29.2	30.7		
		MCWB	21.4	20.9	19.9	17.2	14.1	12.3	11.0	11.4	13.7	16.4	18.5	19.3		
	10%	DB	29.8	28.1	26.2	23.2	19.8	16.5	16.4	17.8	21.3	24.7	26.8	28.1		
		MCWB	20.9	20.2	19.4	16.5	13.5	11.9	10.4	10.8	13.1	15.8	17.8	19.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.8	24.2	22.7	20.1	18.2	15.7	15.5	15.4	17.5	19.8	22.3	22.9		
		MCDB	33.7	32.0	27.3	25.6	21.1	17.2	18.6	20.4	24.9	27.1	30.5	30.9		
	2%	WB	23.2	22.8	21.7	19.1	16.5	14.5	13.6	13.9	16.2	18.6	20.9	21.8		
		MCDB	30.4	28.9	26.5	23.8	19.9	16.4	16.5	18.8	22.3	25.1	28.2	28.7		
	5%	WB	22.4	21.8	20.9	18.2	15.3	13.7	12.3	12.8	15.1	17.7	19.9	20.9		
		MCDB	28.7	26.9	25.6	22.5	19.5	15.9	15.7	17.1	20.6	24.0	26.4	27.0		
	10%	WB	21.6	21.2	20.2	17.5	14.4	13.0	11.3	11.9	14.1	16.9	18.9	20.1		
		MCDB	27.3	26.0	24.5	21.5	18.5	15.3	15.0	16.1	19.3	22.8	23.9	25.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	10.9	10.7	11.7	12.6	10.6	12.7	13.2	13.7	13.4	12.3	11.8		
		MCDBR	17.7	15.7	14.2	14.5	14.7	12.2	15.0	16.4	18.4	19.3	18.0	17.1		
	5% WB	MCWBR	5.4	5.1	5.6	6.0	7.3	6.9	8.1	8.2	7.9	7.6	6.5	5.6		
		MCWBR	13.5	11.7	11.7	12.2	12.1	8.9	12.0	13.3	14.8	14.7	14.8	13.3		
(40) Clear-Sky Solar Irradiance	taub		0.394	0.390	0.360	0.332	0.314	0.305	0.296	0.303	0.336	0.356	0.382	0.399		
		taud	2.430	2.465	2.526	2.576	2.578	2.603	2.609	2.581	2.491	2.463	2.429	2.391		
	Ebn at Noon		943	926	919	891	855	839	869	910	928	949	948	942		
		Edh at Noon	122	114	100	85	74	68	71	82	102	113	122	128		
(44) All-Sky Solar Radiation	RadAvg		6.43	5.47	4.69	3.81	3.02	2.38	2.79	3.68	4.91	5.70	6.04	6.61		
	RadStd		0.50	0.51	0.42	0.33	0.20	0.21	0.21	0.22	0.29	0.52	0.56	0.59		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	+0.44	N/A	N/A	+0.77	N/A	N/A	-2	-83	+182	+94

Nomenclature: See separate page